

Productions of HKIDB's Computerised Mini Braille Press

- ❑ Twelve different types of greeting cards for special occasions such as festivals, birthdays, and so on.
- ❑ A unique all-time desktop calendar of inspirational thoughts.
- ❑ 101 amazing science experiments and brain tickling activities for students.
- ❑ Human anatomy Atlas.
- ❑ An India Atlas titled 'India at your Fingertips'.
- ❑ Books in Braille and large print so that deaf-blind and low-vision students can read them. These include *Who Moved My Cheese*, the Chicken Soup series, *Indian Cookery* and *Light From Many Lamps*.
- ❑ Braille labelling, another upcoming income-generating line.



Attractive educational material in Braille with tactile graphics

hear any instructions. Yet, he does not falter or hesitate while working on the computer.

Dhale is an assistant computer trainer at the Helen Keller Institute for Deaf and Deafblind (HKIDB), Mumbai, a pioneering institute in India and Asia in the education of the deaf-blind. In 1977, founder Beroz Vacha and two other teachers—Diana Fernandes and Brinda Nanavati, started with just three students and a meagre Rs 150. At the time, services existed for the deaf and blind separately, but none were available for the deaf-blind.

In 1979, they moved to the present municipal school at Byculla. HKIDB provides early education services for deaf-blind children, a school for older students, vocational training for rehabilitation and a teacher-training course. Most of the students come from the lower strata of society, but from families with a vision of their children having access to adequate educational facilities and happy futures.

The Legacy Of Helen Keller

Dr Helen Keller (1880-1968) who paved the way for educating the deaf and deaf-blind, was herself deaf-blind. Says Vacha, "I believe that there is no dearth of goodness in this world. Our strength comes from Helen Keller's legacy. Our belief that 'No one is incapable of being educated, loved and accepted with dignity; it is the fundamental birthright of every child to be educated, loved and accepted with dignity' has served us over the years. Our students think, 'we can because we think we can'."

The deaf and deaf-blind have differing degrees of both vision and hearing loss, that could be accompanied by other debilitating conditions such as mental retardation, cerebral palsy and seizures. A deaf-blind child, due to the combination of hearing and visual handicaps, needs a specialised, individualised programme that will help him or her function better. By and large, HKIDB has a 1:1 teacher-student ratio.

HKIDB follows the philosophy of total communication—oral, aural, manual, print, mime and dance that helps improve cognitive levels, thus enabling free modes of self-expression.

Why Teach Computing?

The educational curriculum at HKIDB aims to develop a student's literary and academic skills including reading and writing, cognitive skills (reasoning, attention to tasks, memory, retention, cause and effect), motor skills (such as hand-eye coordination), perceptual skills, orientation and mobility. So far, these skills were taught the old-fashioned way, using books or charts in Braille.

Computers have become a vital part of the modern education process. Some children with deaf-blindness are disinclined to explore their environment—computers motivate such children and improve concentration. Software with attractive visual and auditory features encourage low vision or hearing impaired children to use their residual vision and hearing—larger fonts enable children suffering from low vision to read comfortably.

Vacha says, "How long will the

Computers have become a vital part of the modern education process for the deaf-blind students as well

deaf and deaf-blind earn a living making flowers, candles and paper bags? They are not lesser mortals! They can be as competent as the next person if given the right chance. Don't they have an equal right to be computer-savvy? Deaf-blind children are bright and have tremendous potential for education and work. Computer literacy enables children with disabilities to lead a normal life."

Former students Zamir Dhale and Pradip Sinha (both deaf-blind) insisted on learning computers. Both are now computer training assistants at HKIDB.

The Cybernetic Approach

"Language is the channel through which you become expressive. Language acquisition takes place almost automatically and in due course for people with normal vision and hearing. However, the deaf and deaf-blind must first learn to communicate using language before they can move on to using technology.

"But, how do you expect a deaf-blind person to use the PC or even the typewriter without knowing the alphabet? For that, they must first learn to communicate through sign language, palm-printing, writing and reading in Braille," adds Vacha.

She reflects, "Most suggestions centred on imprinting the tactile Braille script on the keyboard. Again, that would mean undermining their intelligence. I insisted that all the deaf-blind learned touch-typing, a pre-requisite to operate the keyboard. While our aim is to work effectively with electronic technology, we cannot afford to ignore manual technology such as the portable 'Braille' that teaches alphanumeric typing in Braille. We essentially follow a cybernetic approach."

Teaching With Tender Loving Care

Katy Gundevia has been teaching English and other subjects at HKIDB for three years. She says, "My initial reaction when I was faced with the prospect of teaching them was—how can I do it? I learnt palm printing and sign language. On hearing that we would soon use computers, I was instantly relieved! The familiar sight of the PC was encouraging. It is now easier to communicate with students through computers."